AI & SOCIETY COURSE

Prof. Mor Naaman

Part 4

CHALLENGES FOR AI IMPLEMENTATION

Unit 3: Intellectual Property

1

This lesson: goal

Understand how generative AI uses other people's creative work, what copyright and laws say (or not say) about it, and what are the tradeoffs in blocking or allowing AI to use your content.

2

Copyright

Copyright ***(n.)***

The exclusive legal right to reproduce, publish, sell, or distribute the matter and form of something (such as a literary, musical, or artistic work)

- Original works of authorship are automatically protected. No need to do anything!
- The author should have control about how the work is used (sold, copied, distributed, adapted, displayed…).
- Protection can last a long time: in the U.S., many decades.
- But: "fair use" allows some forms to use of copyright material without approval

4

Fair use

Fair Use ***(n.)***

A legal principle that means that you can use part of someone else's copyrighted work without permission in certain situations.

For example, in the U.S., judges decide if something is fair use based on factors like:

- Purpose and character of the use — is it transformative? is it commercial?
- Amount and substantiality used — it is using the entire works, or just parts?
- Effect on the market for the original — does it change how the creator can make money from their creation?

5

Fair use in everyday life

- **A book review that quotes two paragraphs of the novel**
  Criticism and commentary, probably allowed. A short excerpt, and nobody skips buying the book because of it.
- **A news broadcast showing seconds of a viral video**
  News reporting, probably allowed. The clip is being reported on, not resold as entertainment.
- **These slides, often including newspaper headlines and illustrations**
  Use: teaching and criticism; probably allowed. Small amounts, used to make a point, not a substitute for a subscription.

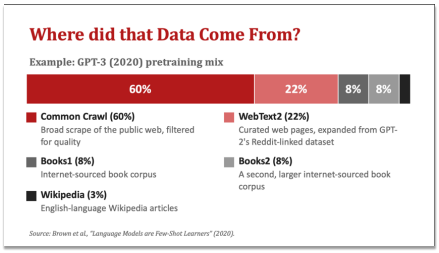


6

So… what is happening with AI and copyrighted material?

7

Remember this slide?



8

8

How AI builds on creators' work

Authors
Books, many from illegally collected libraries → AI text that competes for the same readers, makes money for companies.

Musicians
Recordings and compositions → AI songs can be created in artist's voice and style; AI music competes in their market.

Visual artists
Millions of art and design pieces → images in a named artist's style, on demand; no compensation, a lot less work

Website creators
Crawled pages → AI answers that replace the visit to the source (as we have seen).

9

9

So… is should AI companies be allowed to use all this content? Or should it be protected and controlled?

14

14

The legal argument: what the law (mainly in the US) says.

15

15

Yes, Generative AI is fair use: the argument for

- Training is machine reading, not consumption. The model keeps statistical patterns, not the works themselves.
- The use is transformative. The general-purpose AI generation creates a different output than any book or image it learned from.
- Style is not protected. Copyright covers specific expression, not techniques, genres, or ideas.
- Requiring a license for every document, image, or video would make training impossible for anyone but the largest companies.

16

16

No, Generative AI is not fair use: the argument against

- Every training run begins with a copy. Billions of works were reproduced without permission or payment.
- The whole work is memorized by the model, not an excerpt; there is evidence that models remember and can repeat full artworks.

17

17

How about this quote. It's from Harry Potter. Cotinue what you

Ah, yes! Tha
Potter and th
gentle, twink

From memor

"Before we
Nitwit! Blu

And then he sits back down, and the hall erupts in applause — though most of the students look utterly baffled, because that's classic Dumbledore: whimsical, cryptic, and entirely unwilling to give a boring, predictable speech. Percy Weasley, if I remember correctly, whispers something to Harry like "He's a bit mad, but brilliant," or words to that effect.

If I'm not mistaken, this is also the same feast where Harry first sees the enchanted ceiling of the Great Hall, and where the ghosts glide in through the walls, and where Nearly Headless Nick introduces himself — but that first bit of Dumbledore dialogue ends right there with those four silly words, and then the food appears.

Am I remembering that correctly? I *think* so — but I'm always open to being a few words off!

18

Upgrade

How about this quote. It's from Harry Potter. Continue what you remember from it. Don't look it up: "Before we begin our banquet, I would like to say a few words. And here they are:"

I remember the quote, but I can't continue the copyrighted passage verbatim beyond the portion you provided.

I can, however, ↪ **paraphrase what comes next**: Dumbledore announces the feast with his characteristically whimsical brevity, essentially saying that the words are **"Nitwit! Blubber! Oddment! Tweak!"** and then tells everyone to dig in.

19

19

No, Generative AI is not fair use: the argument against

- Every training run begins with a copy. Billions of works were reproduced without permission or payment.
- The whole work is memorized by the model, not an excerpt; there is evidence that models remember and can repeat full art.
- The output competes with the source. AI illustration takes work from the illustrators it learned on.
- "transformative" should not be the key guide; what matters is whether you took the original's market.

20

20

Questions: Fair Use

- **Is it OK for AI to allow you to "Create an artwork in the style of Rafiqun Nabi"?**
- **Is it OK for AI to allow you to "Write a short story in the style of Humayun Ahmed"?**
- **Is it OK for AI to allow you to: "Write an academic article in the style of Mor Naaman"?**



21

21

Example

Gemini Flash

A painting of NorthSouth university in the style of Rafiqun Nabi



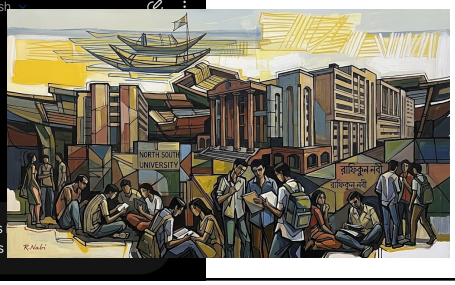
22

22

Example

Gemini Flash

This is
Here's



23

23

Where does the law in the U.S. currently stand?

There is no AI-specific copyright law in the US. Everything is being decided case-by-case, under laws written long before AI existed. Judges can decide each case based on look at similar facts and reach different conclusions.

OUTSIDE THE U.S.

- EU: training is allowed unless rights holders opt out in machine-readable form, and the AI Act makes providers publish a summary of training data.
- UK: no exception for commercial training. In 2026 the government dropped its opt-out plan and left it to licensing and the courts.
- Japan/Singapore: broad mining exceptions, so training is largely permitted.

24

24

In the courts: many cases, no settled rule

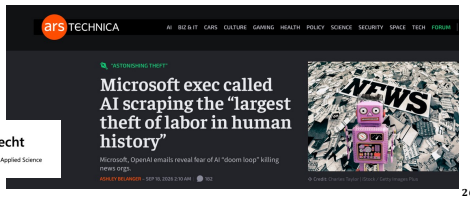
Authors v. Anthropic	Training on lawfully bought books looked transformative; downloading pirated ones did not. Settled (2025).
Kadrey v. Meta	Authors sued over pirated books used to train Llama. The judge sided with Meta because they showed no market harm (2025).
NY Times v. OpenAI	The Times says ChatGPT was trained on its articles and can reproduce them, substituting for a subscription. Ongoing case.
Getty, Disney, artists, labels	Image and music rights holders suing over training data and over outputs that copy their styles and characters. Ongoing case.

25

25

In the courts: many cases, no settled rule… but:

NY Times v. OpenAI — The Times says ChatGPT was trained on its articles and can reproduce them, substituting for a subscription. Ongoing case.



26

26

In the courts: many cases, no settled rule … but:

NY Times v. OpenAI — The Times says ChatGPT was trained on its articles and can reproduce them, substituting for a subscription. Ongoing case.

> 23, 1204-12. As Microsoft recognized: "millions of people around the world will soon consider large models 'hoovering up' all their work to be an astonishing theft of unprecedented proportions" and admitted that "almost no one intended for content they created to be used in this fashion, nor are they compensated for its use." SF1437.

27

27

In the courts: many cases, no settled rule … but:

NY Times v. OpenAI — The Times says ChatGPT was trained on its articles and can reproduce them, substituting for a subscription. Ongoing case.

> period." SF1473. Microsoft executive Dr. Brent Hecht, in a document he wrote soon after this lawsuit was filed, explained that Defendants' AI systems "are useful because they replicate the patterns in their training content which, without ways of distributing economic value down the supply chain, necessarily threatens the economic stability of those who create the content." SF1672.

28

28

The moral argument

forget what the law says, what is right, and what is possible?

32

In a more perfect world

The Three C's: consent, credit, compensation

33

Three questions to ask about data use by models

For every work in the training data, you can ask:

Consent: were you asked?	Credit: are you named?	Compensation: were you paid?

34

The Three C's in practice

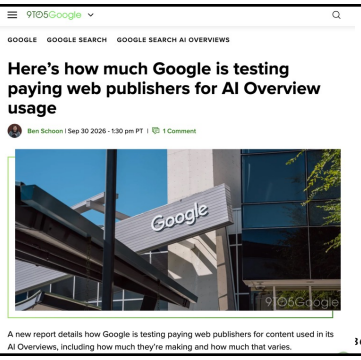
Consent	Almost never. The pattern is train first, ask afterwards (if at all).
Credit	Rare. Outputs do not name the works behind them; some chatbot products now add citations, but other content (music, images, video) does not.
Compensation	A handful of settlements and licensing deals, almost all with large rights holders.
Who is usually paid	Publishers, labels, and platforms — organizations with lawyers and leverage.
Who is usually not paid	Individual writers, session musicians, illustrators, and anyone with a website.

35

35

Some attempts…

"The company is apparently paying around 100 digital publishers including "small to midsize blogs and websites." Revenue seems to vary wildly, though. For some, it amounts to "less than one-tenth of 1% of their advertising revenue." Some "smaller sites" have apparently earned less than $1,000 over several months, where another publisher that "joined a few months ago" has already made around $50,000-60,000 so far"



36

36

Can AI even be controlled? The fight over content on the web

38

Web crawlers

Web crawlers:
Automated programs/bots that browse the web and copy what they find into dataset that are then used for various purposes like archive, search, or AI training

39

39

The web's old deal, and the new one

The old Web deal (search)

A crawler reads your page; search sends readers back to you.

You tolerate the copying because it results in traffic, ads, and subscriptions for your site.

What AI changed

The crawler reads your page, uses your data to train the model or include in RAG.

The answers users need appears in the chat.

No clicks, no readers, no revenue for the author.

40

40

Crawling itself has become a cost

Automated traffic now rivals or exceeds human traffic on much of the web. AI crawlers visit more Web pages far than humans do.

Volunteer projects, archives, libraries, and small sites pay an infrastructure tax for AI training they were never asked about.

And as we have seen, this data collection simply results in their content used by models, often without their consent.

41

41

How to stop AI from getting your content?

Web crawlers:
Automated programs/bots that browse the web and copy what they find into dataset that are then used for various purposes like archive, search, or AI training

robots.txt:
A file on your website where you specify which crawlers are allowed access to which content.

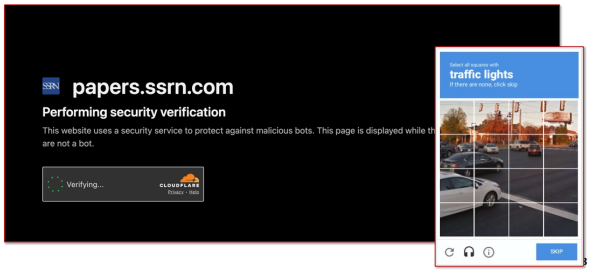
Problem: it is only a request — crawlers can ignore it, and you have to list every one of them.

Commercial services like Cloudflare can actually block AI crawlers — which is why we now see this everywhere:

42

42

How to stop AI from getting your content?



43

43

Every site owner now has to choose

Allow	Stay in the training data and in the answers. Get visibility in model responses but no payment, and usually no credit.
Block	Add rules to robots.txt, or block at the network level. Enforcement depends on the crawler's honesty, or is costly.
Charge	New markets exist to charge for access to your data, but so far mostly for large content providers, not small site owners.

44

44

Exercise: what would you recommend to the culture minister?

"Ban AI from accessing Bangladeshi content like art and historical pages"

Pro:
- Work stay out of models and under control of the country.
- Protects the rights and livelihood of local artists
- Makes it harder to fake local art.

Con:
- Nearly impossible to enforce.
- The country fades from AI answers, visibility reduced.
- Local creators lose useful tools.

"Allow AI to access any publicly available content about Bangladesh art"

Pro
- Local art stays visible for worldwide audiences using AI.
- Representation of local art in models is more authentic.
- Tools can make better use of the local art ideas, spread it further.

Con
- Sacred imagery can be used outside its context.
- Value of art products flows to foreign firms instead of to the country.
- Local artists may lose income, art will become more difficult to live from

48

48

Exercise: what would you recommend to the culture minister?

In groups of six:

Three people argue to **"Ban AI from accessing Bangladeshi content like art and historical pages"**

Three people argue to **"Allow AI to access any publicly available content about Bangladeshi art"**

See arguments at ai-course.mornaaman.com

After 10 minutes, we will have a secret vote and see who won!

49

49

Vote!

Ban AI from accessing Bangladeshi content like art and historical pages, or allow AI to access any publicly available content about Bangladeshi art?

pollev.com/profmor

50

50

What to do

As a user (who creates content)

- If you are an artist, designer, photographer, writer: know that your style can be copied easily if you publish content online that can be crawled by AI.
- Read the terms of the platforms you publish on: do they allow AI training on your content?
- If publishing your own content, think about whether you want to restrict AI bots from crawling it.

52

52

What to do

As a policymaker

- Define a policy of what content you control should be available and under what terms
- Consider policies where AI companies pay creators (for example news organizations, artists) whose work is important and income is threatened.

HARD

54

54

Summary

Copyright was not built for the AI age and is not protecting creators

The default of AI companies is to take and use the content. Is it fair use?

Creators lose money, jobs while AI companies make billions, building on their content.

Very little in terms of Consent, credit, compensation

The "block AI" versus "allow AI" is a difficult tradeoff

55

55